Homework 1 · Problem solutions

Instructor guide · Lectures 1 and 2

Problem 1. Two tickets, one weekend

(a) Rankings and rationality. All three programmes have utility 3: the score pairs are $(6,3)$, $(3,9)$ and $(3,4)$. Thus they are mutually indifferent. A real-valued representation gives completeness and transitivity. The minimum of two continuous linear functions is continuous, so the represented weak-preference graph is closed.

(b) Geometry. The level-6 indifference set is the polyline connecting

$$(0,6), \qquad (12/5,6/5), \qquad (6,0).$$

The weak upper contour set is

$$\{x \geq 0 : 2x_1 + x_2 \geq 6, \quad x_1 + 3x_2 \geq 6\}.$$

For $0 \leq x_1 \leq 12/5$, the indifference set follows $x_2 = 6 - 2x_1$. Mira's score is the smaller one there (strictly smaller before the kink). For $12/5 \leq x_1 \leq 6$, it follows $x_2 = (6 - x_1)/3$, and Anton's score is the smaller one. At the kink both scores are 6. The upper contour is above both lines, not a union of the two half-spaces.

(c) Properties. At any reference utility level $k \geq 0$, the upper contour is the intersection of X with two half-spaces $2x_1 + x_2 \geq k$ and $x_1 + 3x_2 \geq k$, so preferences are convex. They are not strictly convex: $x = (0,4)$ and $y = (1,2)$ both have utility 4, and their midpoint $(1/2,3)$ also has utility 4.

Both linear scores have strictly positive coefficients. If a bundle increases coordinatewise with at least one strict improvement, *both* scores strictly rise. More precisely, both score increases have a strictly positive minimum, so their minimum also rises strictly. This proves strong monotonicity. Increasing either coordinate by an arbitrarily small positive amount then proves LNS throughout the orthant.

(d) Displayed scores. Since $u(X) = [0,\infty)$, squaring is strictly increasing on the attained range and preserves every comparison. But v is not concave. On $x_2 = 0$, $u(x) = x_1$ and $v(x) = x_1^2$. For example,

$$v(1,0) = 1 < \tfrac{1}{2}v(0,0) + \tfrac{1}{2}v(2,0) = 2.$$

Changing the numerical scale need not preserve concavity.

(e) Convex representation. Set $x = (0,6)$ and $y = (6,0)$. Both have utility 6, while their midpoint $(3,3)$ has utility 9. Any representation g must satisfy $g(x) = g(y) < g((x+y)/2)$, whereas convexity would require $g((x+y)/2) \leq [g(x)+g(y)]/2 = g(x)$. Thus no convex representing function exists. Convexity of preferences does not imply convexity of a utility index.

Problem 2. One more lesson, one less coffee

(a) Existence, exhaustion and uniqueness. The budget is nonempty and compact for $p > 0, m > 0$, and utility is continuous, so an optimum exists. Utility is strictly increasing in a and q, hence an optimum exhausts the budget. Substitute $a = m - pq$ to obtain

$$\max_{0 \le q \le m/p} \{m - pq + 6\log(1+q)\}.$$

The second derivative with respect to q is $-6/(1+q)^2 < 0$. The reduced problem has a unique maximiser; budget exhaustion then uniquely determines a. The original utility is not strictly concave in the pair (a, q) because it is affine in a; do not use that false premise to justify uniqueness.

(b) Kuhn–Tucker system. With constraints written as nonnegative,

$$\mathcal{L} = a + 6\log(1+q) + \lambda(m - a - pq) + \mu_a a + \mu_q q.$$

Stationarity:

$$1 - \lambda + \mu_a = 0, \qquad \frac{6}{1+q} - \lambda p + \mu_q = 0.$$

Primal feasibility: $a, q \ge 0$, $m - a - pq \ge 0$. Dual feasibility: $\lambda, \mu_a, \mu_q \ge 0$. Complementary slackness:

$$\lambda(m - a - pq) = 0, \qquad \mu_a a = 0, \qquad \mu_q q = 0.$$

The objective is concave and differentiable on a neighbourhood of the consumption domain; the constraints are affine. KKT candidates satisfying these conditions are globally optimal. The reduced problem in (a) already establishes uniqueness.

(c) Demand. The reduced derivative is $-p + 6/(1+q)$, strictly decreasing in q. Thus

$$q^*(p,m) = \min\left\{\frac{m}{p},\ \max\left\{\frac{6}{p} - 1, 0\right\}\right\}, \qquad a^*(p,m) = m - pq^*(p,m).$$

The full classification is

$$(a^*, q^*) = \begin{cases} (m, 0), & p \ge 6, \\ (0, m/p), & 0 < p < 6, \quad 0 < m \le 6 - p, \\ (m - 6 + p, 6/p - 1), & 0 < p < 6, \quad m > 6 - p. \end{cases}$$

At $p = 6$, demand for tutoring is still uniquely zero: the reduced derivative is zero only at zero and negative thereafter. At $m = 6 - p$ with $p < 6$, $a = 0$ and $q = 6/p - 1 = m/p$; this is the boundary between the two positive-tutoring regimes, not a multiplicity of optima.

The corresponding multipliers are

$$\lambda = \begin{cases} 1, & p \ge 6, \\ 6/(p+m), & 0 < p < 6, \quad m \le 6 - p, \\ 1, & 0 < p < 6, \quad m > 6 - p, \end{cases} \qquad \mu_a = \lambda - 1.$$

When $q > 0$, $\mu_q = 0$; when $q = 0$, $\mu_q = p - 6 \ge 0$. These formulas agree on the boundaries. In particular, a binding zero lower bound on a good need not have a strictly positive multiplier.

(d) Two incomes. At $p = 2$,

m	a^*	q^*	λ
2	0	1	$3/2$
8	4	2	1

With $m = 2$, the no-borrowing/nonnegativity constraint on a binds; the student would prefer to buy more tutoring at these prices but cannot finance it. For $m < 4$, each additional unit of income buys $1/2$ more tutoring hour. With $m = 8$, tutoring is at its unconstrained preferred level of 2 hours and additional income goes to a. Quasilinearity does not eliminate low-income corner effects.

(e) Indirect utility and Roy. In the open region $0 < p < 6$, $m > 6 - p$, both goods are consumed and

$$v(p, m) = m - 6 + p + 6\log(6/p), \qquad v_m = 1, \qquad v_p = 1 - \frac{6}{p}.$$

Hence $-v_p/v_m = 6/p - 1 = q^*$. The derivative v_m is the marginal value of income in the chosen utility units, equal to the budget multiplier. In this region one extra unit of income buys one more unit of a and raises utility by 1. The multiplier is not invariant to rescaling the utility representation.